CSE 270 Sports Analytics

Basketball Individual performance

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NBA Individual performance

NBA Individual performance

2 main stages of the game.

- Attacking
- Defending

https://www.youtube.com/watch?v=wYjp2zoqQrs

NBA Individual performance

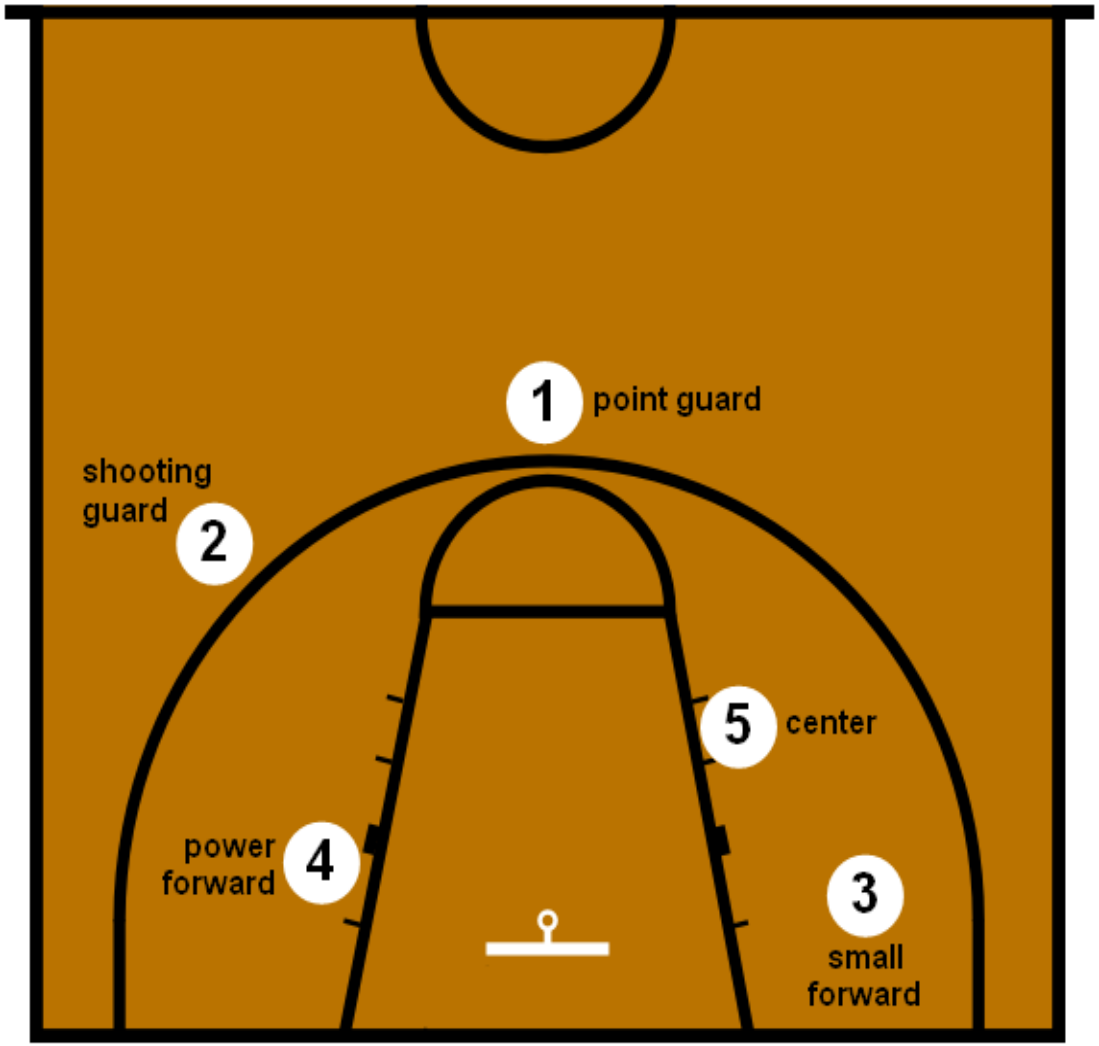
5 main positions on the field.

- Center - On offense, the center tries to score on close shots and rebound. But on defense, the center tries to block opponents' shots and rebound their misses.
- Power Forward - The power forward does many of the things a center does, playing near the basket while rebounding and defending taller players. But power forwards also take longer shots than centers.
- Small Forward - The small forward plays against small and large players. They roam all over on the court. Small forwards can score from long shots and close ones.
- Point Guard - The point guard runs the offense.The point guard defends the opponent's point guard and tries to steal the ball.
- Shooting Guard - The shooting guard can make shots from long distance and also is a good dribbler.

Let's watch a short video,

https://www.youtube.com/watch?v=4_4CymXARWQ

NBA Individual performance



NBA Individual performance

Defining attributes for each position.

- Center - The center is the tallest player on each team, playing near the basket.
- Power Forward - The power forward is very close to center player, but in this position they should also have good shooting attributes.
- Small Forward - The small forward should have good shooting efficiency in close and far distances.
- Point Guard - The point guard should have good defending characteristics.
- Shooting Guard - The shooting guard should have good dribbling attributes and shooting accuracy in long ranges.

What measures other than the height and shooting can be used to position the players on the field ?

NBA Individual performance

In game statistics

- STEALS
- TURNOVERS - Turnover occurs when a team loses possession of the ball to the opposing team before a player takes a shot at their team's basket.
  https://www.youtube.com/shorts/I4MDXOWrUmk
- REBOUNDS
  1. OFFENSIVE REBOUNDS - An offensive rebound occurs when offensive players regain possession of the ball after a missed shot.
  2. DEFENSIVE REBOUNDS - Defensive rebounds occur when defensive players gain possession of the ball after an offensive player misses the shot.
- BLOCKS - Block occurs when a defensive player legally deflects a field goal attempt from an offensive player to prevent a score.
- FOULS
  1. PERSONAL FOULS
  2. FLAGRANT FOULS
  3. TECHINCAL FOULS

NBA Individual performance

Dataset

```
str(nba_players)
```

```
## 'data.frame':    4658 obs. of  34 variables:
##  $ League     : chr  "NBA" "NBA" "NBA" "NBA" ...
##  $ Season     : chr  "1999 - 2000" "1999 - 2000" "1999 - 2000" "1999 - 2000" ...
##  $ Stage      : chr  "Regular_Season" "Regular_Season" "Regular_Season" "Regular_Season" ...
##  $ Player     : chr  "Shaquille O'Neal" "Vince Carter" "Karl Malone" "Allen Iverson" ...
##  $ Team       : chr  "LAL" "TOR" "UTA" "PHI" ...
##  $ GP         : int  79 82 82 70 82 82 74 81 82 75 ...
##  $ MIN        : num  3163 3126 2947 2853 3425 ...
##  $ FGM        : int  956 788 752 729 747 619 696 759 748 748 ...
##  $ FGA        : int  1665 1696 1476 1733 1666 1447 1422 1526 1636 1548 ...
##  $ X3PM       : int  0 95 2 89 177 83 34 30 99 27 ...
##  $ X3PA       : int  1 236 8 261 520 288 98 81 247 95 ...
##  $ FTM        : int  432 436 589 442 311 618 480 309 260 311 ...
##  $ FTA        : int  824 551 739 620 423 758 604 404 317 414 ...
##  $ TOV        : int  223 178 231 230 224 311 240 268 196 218 ...
##  $ PF         : int  255 263 229 162 178 188 190 205 171 264 ...
##  $ ORB        : int  336 150 169 71 100 118 97 223 122 189 ...
##  $ DRB        : int  742 326 610 196 429 197 393 733 396 598 ...
##  $ REB        : int  1078 476 779 267 529 315 490 956 518 787 ...
##  $ AST        : int  299 322 304 328 732 365 385 401 438 345 ...
##  $ STL        : int  36 110 79 144 153 103 103 120 109 120 ...
##  $ BLK        : int  239 92 71 5 18 36 43 126 32 128 ...
##  $ PTS        : int  2344 2107 2095 1989 1982 1939 1906 1857 1855 1834 ...
##  $ birth_year : int  1972 1977 1963 1975 1968 1974 1972 1976 1973 1973 ...
##  $ birth_month: chr  "Mar" "Jan" "Jul" "Jun" ...
##  $ birth_date : chr  "Mar 6, 1972" "Jan 26, 1977" "Jul 24, 1963" "Jun 7, 1975" ...
##  $ height     : chr  "7-1" "6-6" "6-9" "6-0" ...
##  $ height_cm  : int  216 198 206 183 193 198 203 211 201 206 ...
##  $ weight     : chr  "325" "220" "265" "165" ...
##  $ weight_kg  : int  147 100 120 75 82 99 102 109 102 111 ...
##  $ nationality: chr  "United States" "United States" "United States" "United States" ...
##  $ high_school: chr  "Robert G. Cole High School " "Mainland High School" "Summerfield High School" "Bethel High School " ...
##  $ draft_round: int  1 1 1 1 1 1 1 1 1 1
```

NBA Individual performance

Let's select the performance metrics.

```
nba_players <- nba_players %>%
  select(Season, Player, height_cm, weight_kg,
         GP, MIN, STL, TOV, ORB, DRB, REB, PF, BLK, PTS, AST,
         FTM, FTA, FGM, FGA, X3PM, X3PA, ) %>%
  arrange(Player, Season)  %>%
  mutate(PR = STL - TOV)
head(nba_players, 3)
```

```
##        Season       Player height_cm weight_kg GP    MIN STL TOV ORB DRB REB
## 1 1999 - 2000   A.C. Green       206       100 82 1929.0  53  53 160 326 486
## 2 2000 - 2001   A.C. Green       206       100 82 1411.0  30  45 107 206 313
## 3 2008 - 2009 Aaron Brooks       183        73 80 1998.1  46 125  33 124 157
##    PF BLK PTS AST FTM FTA FGM FGA X3PM X3PA  PR
## 1 127  18 413  80  66  95 173 387    1    4   0
## 2 119   8 367  39  79 111 144 324    0    6 -15
## 3 152   8 894 238 149 172 316 783  113  309 -79
```

NBA Individual performance

Let's try to evaluate the player's performance based on the amount of time they spent on the field.
We will use the players' historical data to measure their performance for each season, by taking 5 preceding seasons only in order to avoid one season stars. We will use success/failure approach by dividing the metrics of the players into successful and non-succesfull actions and then compare their success rate with their playing minutes.

NBA Individual performance

Let's start with defending attributes.

Successful actions, when defending

- STEALS
- BLOCKS
- DEFENSIVE REBOUNDS

Non Successful actions, when defending

- PERSONAL FOULS
- TURNOVER

NBA Individual performance

Let's calculate the difference between the cumulative success actions and cumulative failure actions.

```
nba_players_defense <- nba_players %>%
  arrange(Player, Season)  %>%
  group_by(Player) %>%
  mutate(CUM_STL = rollmean(STL, k = 5, fill = NA, align="right"),
         CUM_TOV = rollmean(TOV, k = 5, fill = NA, align="right"),
         CUM_DRB = rollmean(DRB, k = 5, fill = NA, align="right"),
         CUM_PF = rollmean(PF, k = 5, fill = NA, align="right"),
         CUM_BLK = rollmean(BLK, k = 5, fill = NA, align="right"),
         CUM_MIN = rollmean(MIN, k = 5, fill = NA, align = "right"),
         CUM_SUCCESS_1 = CUM_STL - CUM_TOV,
         CUM_SUCCESS_2 = CUM_STL - CUM_PF,
         CUM_SUCCESS_3 = CUM_DRB - CUM_TOV,
         CUM_SUCCESS_4 = CUM_DRB - CUM_PF,
         CUM_SUCCESS_5 = CUM_BLK - CUM_TOV,
         CUM_SUCCESS_6 = CUM_BLK - CUM_PF,
         SUCCESS_1 = STL - TOV,
         SUCCESS_2 = STL - PF,
         SUCCESS_3 = DRB - TOV,
         SUCCESS_4 = DRB - PF,
         SUCCESS_5 = BLK - TOV,
         SUCCESS_6 = BLK - PF,
         ) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual performance

Let's check if the cumulative metrics are correlated to the amount of time the players spent on the field.

```
cor_with_min <- function(x){
  return(cor(x, nba_players_defense$MIN))
}

nba_players_defense %>%
  select(CUM_SUCCESS_1, CUM_SUCCESS_2, CUM_SUCCESS_3,
         CUM_SUCCESS_4, CUM_SUCCESS_5, CUM_SUCCESS_6, CUM_MIN) %>%
  apply(2,cor_with_min)
```

```
## CUM_SUCCESS_1 CUM_SUCCESS_2 CUM_SUCCESS_3 CUM_SUCCESS_4 CUM_SUCCESS_5
##   -0.37674687    0.05995169    0.11593273    0.28699436   -0.38554778
## CUM_SUCCESS_6       CUM_MIN
##   -0.14944537    0.77526473
```

NBA Individual performance

Now let's do the same for seasonal metrics.

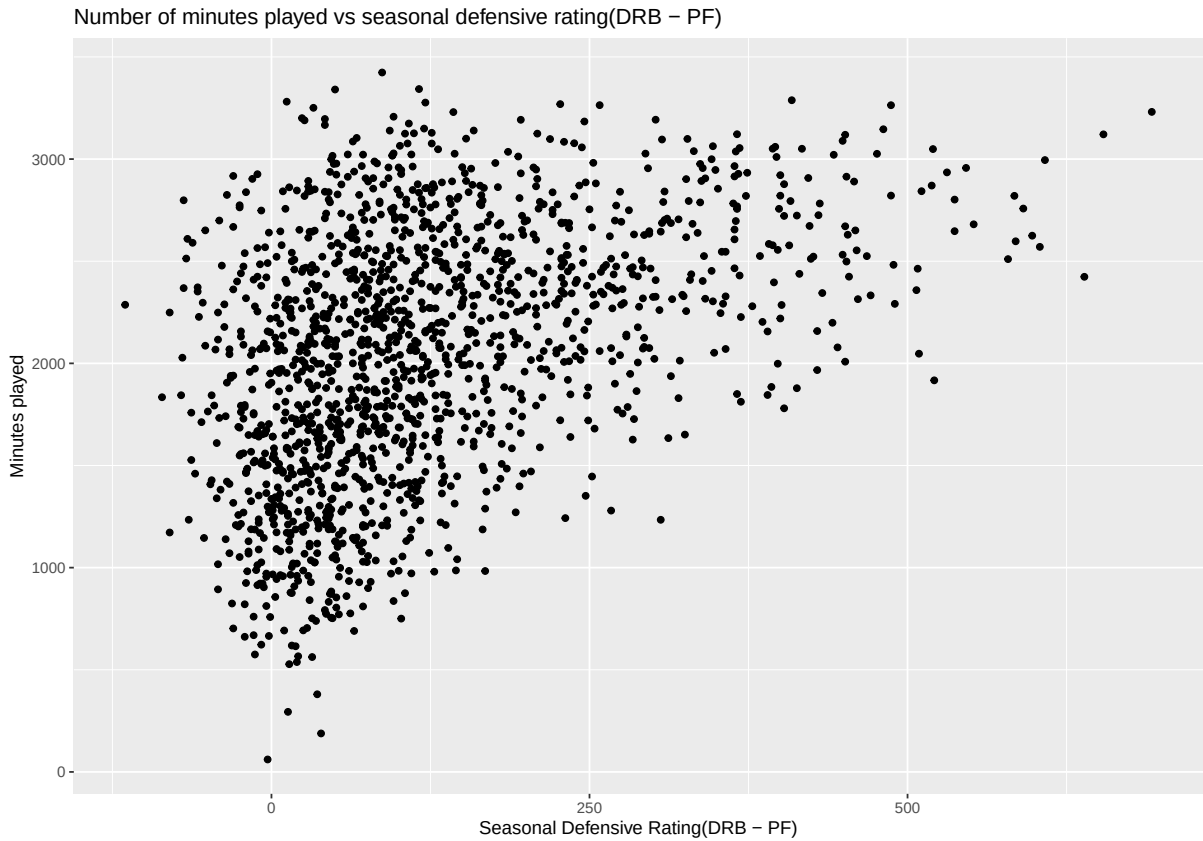
```
nba_players_defense %>%
  select(SUCCESS_1, SUCCESS_2, SUCCESS_3,
         SUCCESS_4, SUCCESS_5, SUCCESS_6) %>%
  apply(2,cor_with_min)
```

```
##  SUCCESS_1  SUCCESS_2  SUCCESS_3  SUCCESS_4  SUCCESS_5  SUCCESS_6
## -0.4841751 -0.1157084  0.2796786  0.4340337 -0.5171269 -0.3671302
```

NBA Individual performance

The metric with the highest positive correlation rate is the seasonal difference between the cumulative number of defensive rebounds made and cumulative number of turnovers made.

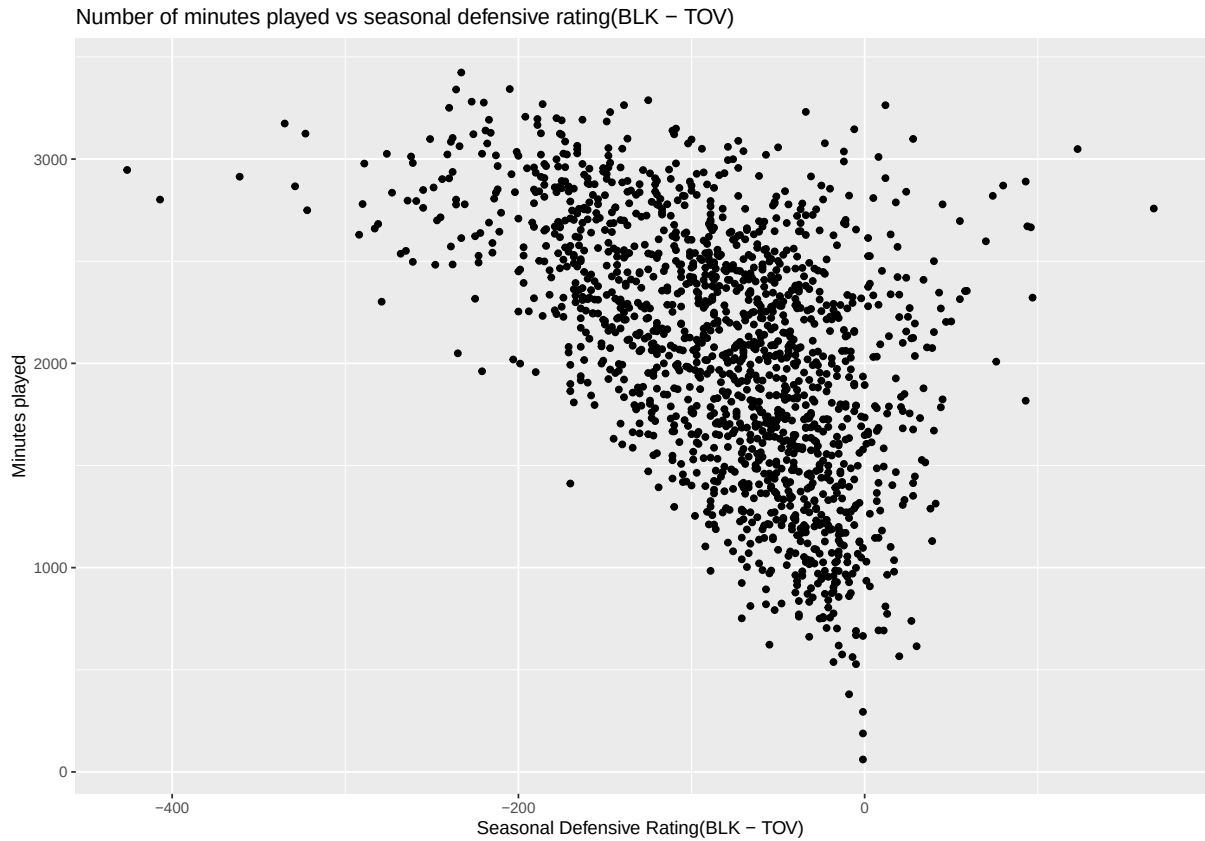
```
ggplot(nba_players_defense, aes(x = SUCCESS_4, y = (MIN))) +
  geom_point() +
  ggtitle("Number of minutes played vs seasonal defensive rating(DRB - PF)") +
  xlab("Seasonal Defensive Rating(DRB - PF)") +
  ylab("Minutes played")
```



NBA Individual performance

The metric with the lowest positive correlation rate is the difference between the cumulative number of blocks made and cumulative number of turnovers made.

```
ggplot(nba_players_defense, aes(x = SUCCESS_5, y = (MIN))) +
  geom_point() +
  ggtitle("Number of minutes played vs seasonal defensive rating(BLK - TOV)") +
  xlab("Seasonal Defensive Rating(BLK - TOV)") +
  ylab("Minutes played")
```



NBA Individual performance

Let's drop the difference between number of steals made and fouls made, as it has almost 0 correlation with the number of minutes played and then fit a model with the defensive power ratings.

```
nba_players_defense <- nba_players_defense %>%
  select(-CUM_SUCCESS_2, SUCCESS_2)

summary(lm(MIN ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5
                  + SUCCESS_5 + CUM_MIN ), data = nba_players_defense))
```

```
##
## Call:
## lm(formula = MIN ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5 +
##     SUCCESS_5 + CUM_MIN), data = nba_players_defense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1055.44  -181.99     5.73   186.91   999.78
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -248.44326   39.43534   -6.30 3.87e-10 ***
## SUCCESS_4        2.91258    0.13177   22.10  < 2e-16 ***
## CUM_SUCCESS_4   -3.12429    0.14333  -21.80  < 2e-16 ***
## CUM_SUCCESS_5    5.07573    0.26602   19.08  < 2e-16 ***
## SUCCESS_5       -5.20481    0.25031  -20.79  < 2e-16 ***
## CUM_MIN          1.06011    0.02223   47.69  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 283.2 on 1541 degrees of freedom
## Multiple R-squared:  0.7874, Adjusted R-squared:  0.7867
## F-statistic:  1141 on 5 and 1541 DF,  p-value: < 2.2e-16
```

NBA Indvidual performance.

Dropping the cumulative average of minutes played to understand the model's power on estimating the players' number of minutes on the field without preceding 5 season.

```
summary(lm(MIN ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5
                  + SUCCESS_5 ), data = nba_players_defense))
```

```
##
## Call:
## lm(formula = MIN ~ (SUCCESS_4 + CUM_SUCCESS_4 + CUM_SUCCESS_5 +
##     SUCCESS_5), data = nba_players_defense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1934.89  -309.24    -8.39   310.76  1395.67
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1491.5602    23.5448  63.350  < 2e-16 ***
## SUCCESS_4        3.0387     0.2072  14.663  < 2e-16 ***
## CUM_SUCCESS_4   -1.1806     0.2162  -5.462 5.49e-08 ***
## CUM_SUCCESS_5    1.2938     0.3994   3.239  0.00123 **
## SUCCESS_5       -5.5845     0.3935 -14.190  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 445.5 on 1542 degrees of freedom
## Multiple R-squared:  0.4735, Adjusted R-squared:  0.4721
## F-statistic: 346.7 on 4 and 1542 DF,  p-value: < 2.2e-16
```

NBA Individual performance

We can see that all the cumulative metrics of the player's except the average number of minutes played in preceeding 5 seaons, do not have as much significance as seasonal values, so we can conclude that the number of minutes the players will have on the field is connected to their previous season statistics, but can change based on their seasonal performance.

NBA Individual performance

For the attacking attributes we will calculate the difference between the attempts made for successful actions and actual successful actions.

Successful actions, when attacking

- FIELD GOALS
- 3 POINT GOALS
- FREE THROWS
- OFFENSIVE REBOUNDS
- POINTS SCORED
- ASSISTS MADE

Non Successful actions, when attacking

- FIELD GOALS ATTEMPTS
- 3 POINT ATTEMPTS
- FREE THROW ATTEMPTS

NBA Individual performance

```
nba_players_offense <- nba_players %>%
  arrange(Player, Season)  %>%
  group_by(Player) %>%
  mutate(CUM_FGM = rollmean(FGM, k = 5, fill = NA, align="right"),
         CUM_FGA = rollmean(FGA, k = 5, fill = NA, align="right"),
         CUM_X3PM = rollmean(X3PM, k = 5, fill = NA, align="right"),
         CUM_X3PA = rollmean(X3PA, k = 5, fill = NA, align="right"),
         CUM_FTM = rollmean(FTM, k = 5, fill = NA, align="right"),
         CUM_FTA = rollmean(FTA, k = 5, fill = NA, align="right"),
         CUM_AST = rollmean(AST, k = 5, fill = NA, align="right"),
         CUM_PTS = rollmean(PTS, k = 5, fill = NA, align="right"),
         CUM_ORB = rollmean(PTS, k = 5, fill = NA, align="right"),
         CUM_MIN = rollmean(MIN, k = 5, fill = NA, align="right"),
         SUCCESS_1 = FGA - FGM,
         SUCCESS_2 = X3PA - X3PM,
         SUCCESS_3 = FTA - FTM,
         CUM_SUCCESS_1 = CUM_FGA - CUM_FGM,
         CUM_SUCCESS_2 = CUM_X3PA - CUM_X3PM,
         CUM_SUCCESS_3 = CUM_FTA - CUM_FTM,
         ) %>%
  drop_na() %>%
  ungroup()
```

NBA Individual performance

Let's check if the cumulative metrics are correlated to the amount of time the players spent on the field.

```
cor_with_min <- function(x){
  return(cor(x, nba_players_offense$MIN))
}

nba_players_offense %>%
  select(CUM_SUCCESS_1, CUM_SUCCESS_2, CUM_SUCCESS_3, CUM_AST, CUM_MIN, CUM_PTS) %>%
  apply(2,cor_with_min)
```

```
## CUM_SUCCESS_1 CUM_SUCCESS_2 CUM_SUCCESS_3       CUM_AST       CUM_MIN
##     0.6183725     0.2545079     0.3706672     0.3978890     0.7752647
##       CUM_PTS
##     0.6307494
```

NBA Individual performance

Let's check if the seasonal metrics are correlated to the amount of time the players spent on the field.

```
cor_with_min <- function(x){
  return(cor(x, nba_players_offense$MIN))
}

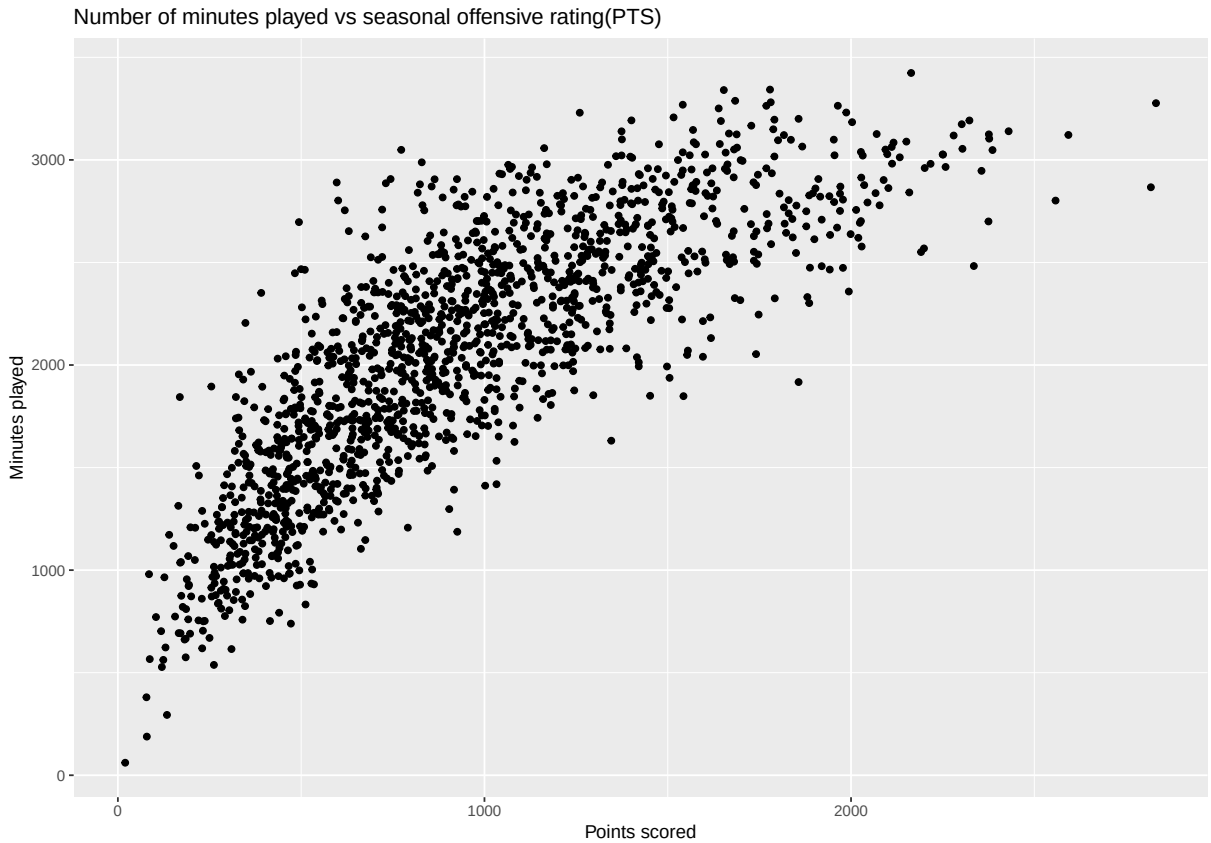
nba_players_offense %>%
  select(SUCCESS_1, SUCCESS_2, SUCCESS_3, AST, PTS) %>%
  apply(2,cor_with_min)
```

```
## SUCCESS_1 SUCCESS_2 SUCCESS_3       AST       PTS
## 0.8062646 0.4003769 0.4892539 0.5565007 0.8085913
```

NBA Individual performance

The seasonal metric with the highest correlation is the difference between the number of points scored by the player.

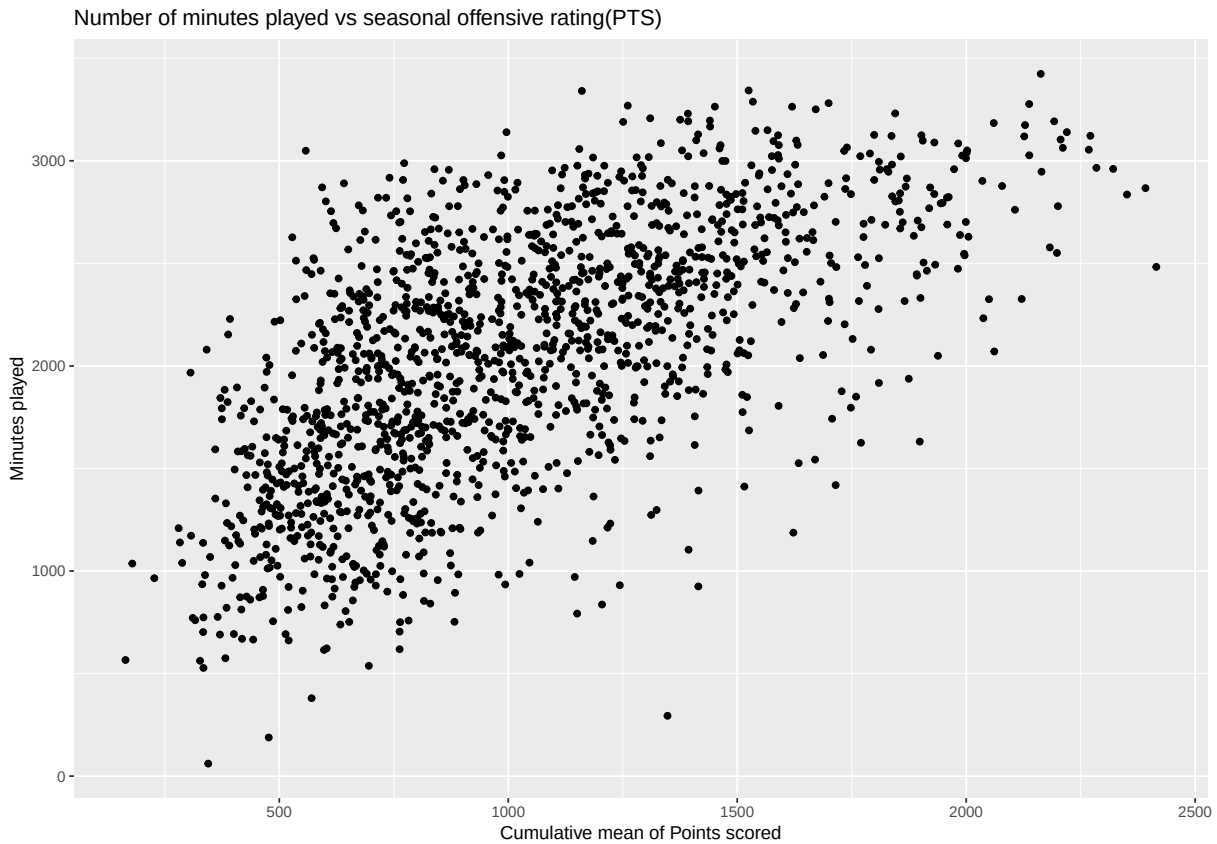
```
ggplot(nba_players_offense,aes(x = PTS, y = MIN)) +
  geom_point() +
  ggtitle("Number of minutes played vs seasonal offensive rating(PTS)") +
  xlab("Points scored") +
  ylab("Minutes played")
```



NBA Individual performance

The cumulative metric with the highest correlation is again the number of points scored.

```
ggplot(nba_players_offense,aes(x = CUM_PTS, y = MIN)) +
  geom_point() +
  ggtitle("Number of minutes played vs seasonal offensive rating(PTS)") +
  xlab("Cumulative mean of Points scored") +
  ylab("Minutes played")
```



NBA Individual performance

```
summary(lm(MIN ~ (PTS + AST + SUCCESS_1 +
                  + CUM_SUCCESS_1 + CUM_PTS + CUM_AST + CUM_MIN ), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MIN ~ (PTS + AST + SUCCESS_1 + +CUM_SUCCESS_1 +
##     CUM_PTS + CUM_AST + CUM_MIN), data = nba_players_offense)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -769.39 -130.14    1.88  132.44  740.66
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -154.98178   28.69808  -5.400 7.69e-08 ***
## PTS              0.80381    0.05673  14.170  < 2e-16 ***
## AST              1.06124    0.09989  10.624  < 2e-16 ***
## SUCCESS_1        0.93755    0.13657   6.865 9.62e-12 ***
## CUM_SUCCESS_1   -0.49817    0.15034  -3.314 0.000943 ***
## CUM_PTS         -0.98863    0.06365 -15.533  < 2e-16 ***
## CUM_AST         -1.04549    0.09988 -10.468  < 2e-16 ***
## CUM_MIN          1.04056    0.01970  52.830  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 199 on 1539 degrees of freedom
## Multiple R-squared:  0.8952, Adjusted R-squared:  0.8947
## F-statistic:  1877 on 7 and 1539 DF,  p-value: < 2.2e-16
```

NBA Indvidual performance.

Dropping the cumulative average of minutes played to understand the model's power on estimating the players' number of minutes on the field without preceding 5 seasons.

```
summary(lm(MIN ~ (PTS + AST + SUCCESS_1 +
                  + CUM_SUCCESS_1 + CUM_PTS + CUM_AST), data = nba_players_offense))
```

```
##
## Call:
## lm(formula = MIN ~ (PTS + AST + SUCCESS_1 + +CUM_SUCCESS_1 +
##     CUM_PTS + CUM_AST), data = nba_players_offense)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1369.01  -238.78    -4.58   219.23  1153.29
##
## Coefficients:
##                 Estimate Std. Error t value Pr(>|t|)
## (Intercept)   1169.27285   23.43170  49.901  < 2e-16 ***
## PTS              0.71444    0.09508   7.514 9.65e-14 ***
## AST              1.05962    0.16750   6.326 3.29e-10 ***
## SUCCESS_1        1.17056    0.22889   5.114 3.55e-07 ***
## CUM_SUCCESS_1    0.05757    0.25148   0.229  0.81896
## CUM_PTS         -0.34395    0.10475  -3.284  0.00105 **
## CUM_AST         -0.68634    0.16709  -4.108 4.21e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 333.7 on 1540 degrees of freedom
## Multiple R-squared:  0.7051, Adjusted R-squared:  0.7039
## F-statistic: 613.6 on 6 and 1540 DF,  p-value: < 2.2e-16
```

NBA Individual performance

We can again see that the seasonal metrics have higher contribution to the number of minutes played, and we can conclude that the players who are making more attempts to score field goal are generally playing more.